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An Explainable Machine Learning Framework for Energy Forecasting, Anomaly Detection, and Cost-Aware Decision Support in Smart Campuses

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Abstract—A consumption forecast does not indicate to a university facilities department what to shut off, what the wastage would cost, or which repair needs to be prioritized. The body of building energy research literature has developed each step of analysis in isolation; the combination of these into a decision process is an area that has received much less attention. This paper proposes CampusWatt, a framework where the forecast is used as a reference point rather than as a solution: the forecast defines the consumption explained by the operating situation, the residual identifies what it does not explain, a review of equipment inventory translates the residual into a list of physically feasible causes, and a high-tariff model assigns a cost to it so competing solutions can be compared financially. Data comes in three streams: bulk upload, daily recording and hourly recording with a deterministic strategy for reconciliation where source of information and any discrepancy becomes an explicit conflict. The evaluation will use a synthesized year of hourly consumption for a five-zone academic building. The zone level gradient boosting model attains a 5.6 per cent MAPE on computer laboratories, compared to a baseline of 72.0 per cent, which has been recalculated using a seasonal-naïve approach. The zone level gradient boosting model outperforms the baseline in each zone. For the detection task, eleven labeled events were used as input for the reference predictor, with nine out of eleven events being detected, while the hour-level precision remained low for all thresholds tested, as explicitly mentioned since the detection experiment defines the reference predictor and not the trained models. Tariff Modelling assigns 50.4 per cent of the annual cost to the maximum-demand tariff. The data used is generated by an audited inventory of appliances and no metered validation was carried out.

Keywords—building energy forecasting, explainable artificial intelligence, anomaly detection, gradient boosting, maximum demand charge, decision support, smart campus

I. INTRODUCTION

One of the largest ongoing cost items borne by an Indian school is electricity and one of the most uninformed. Several buildings are typically fed by a single high-tension supply and billed on the basis of one meter reading per month. This is a gross reading. It tells you only how much electricity has been used, not which part of the building used it or at what hour or for what purpose.

The time lag makes the problem worse. When the bill comes in, the behavior pattern that caused the bill has persisted for four to six weeks. An air conditioner keeps running over vacation, a compressor whose efficiency is waning, a laboratory whose machines never rest all of these stay silent. They are included in a monthly bill that is not remarkable in relation to the preceding month's bill.

Forecasting of building-energy use has been explored extensively; statistical and machine learning approaches have been reviewed in various literature sources [1], [3], [4]. On the contrary, decision-making support following the forecast is poorly developed. It does not matter if the facility manager is informed about an anomaly on Tuesday night because it will remain unclear where the power consumption comes from, how expensive the problem is, and if solving it is more beneficial than other three problems for which the maintenance budget is vying.

A. Problem Statement

Campus energy management takes place regardless of the necessary information. The consumption is tracked at a level of granularity insufficient for pinpointing wasteful behavior, reported with a delay that prevents correction and priced in a manner too simplistic for ranking of possible actions.

Secondarily, there is a limit to what can be achieved in reality. Sub-metering may not be present, and any data that is available comes intermittently; an occasional export from the utility website, coupled with manual entries from the meter into a register by a company employee. The design which requires a constant stream of instrumentation data cannot be used in its absence; one that relies on only manual data cannot leverage what history has been recorded. Both have to be accommodated and reconciled if they conflict.

B. Research Objectives

- O1 — Forecast zone-level consumption accurately enough for use as a detection reference, with only weather and calendar information as input.
- O2 — Trace a deviation from the forecast back to installed equipment, based on an audited inventory without sub-metering.
- O3 — See whether the difference between modelling the whole high-tension tariff versus a simple energy price impacts the ranking of interventions.
- O4 — Reconcile readings obtained via independent channels and differing levels of granularity, preserving provenance and highlighting inconsistencies.
- O5 — Benchmark the quality of the forecast against a reproducible seasonal-naïve reference, and whether labelled injected faults are surfaced by the pipeline.

C. Importance

The issue of institutional energy savings is often discussed in terms of a capital problem, and such actions

have to compete for funds that do not even exist. The current research tries to find out how it is possible to reveal such operations without capital investments, as well as make consumption measurable to understand the reasons behind them. The limitations involved a big facility, only one meter and no budget for analytics are characteristic for institutions that have not had any submetering done.

D. Contributions

The contribution is system level integration and operationalisation rather than a new learning algorithm. Four elements are claimed.

- A reconciliation policy for heterogeneous energy inputs assigning each zone-day one of five provenance states and escalating disagreement beyond a defined tolerance to a human decision.
- An asymmetric validation rule rejecting physically impossible typed values while admitting the same values from imported files, since the second may be the fault the system exists to detect.
- An inventory-bound attribution procedure converting a residual in kilowatts into an enumerated set of physically admissible equipment hypotheses, eliminating those requiring more units than are installed.
- An evaluation using labelled synthetic faults, permitting event-level detection to be measured against ground truth in a setting where metered anomaly labels do not exist.

II. LITERATURE REVIEW AND RESEARCH GAP

A. Forecasting, Detection and Explainability

There have been several surveys conducted on statistical learning load forecasting [1]. Metaheuristic optimization was used for extreme learning machines by Gao et al., and they obtained improvements over the conventional versions [2]; Liu et al. evaluated nine models and fifty-two building features, where they showed that the accuracy depends on the feature construction and not the model used [3]; it provided guidance on the feature design of Section III-E. The campus-specific literature includes occupancy and weather prediction for a university building [5], and multi-year meter data sets [6]. This project uses the standard method but without any enhancements and the forecasts from this process are used as the detection reference and not the product.

Regarding detection, Himeuretal. found prediction-residual methods to be the prevailing paradigm

[7], and Araya et al. proposed an ensemble approach [8]. Telila et al. used a threshold on prediction errors using smart meter readings [9]; their use of induced anomalies to label the records is the method used herein. These methods yield a flagged period, and in this case study, the process of transforming the period into an equipment suggestion is included. As for explainability, Noorchenarboo and Grolinger constrained SHAP background set with each anomaly [10], Moon et al. used Shapley attribute in educational buildings [11], and Chung and Liu compared LIME with SHAP [12]. They all explain the model based on its inputs; a maintenance worker requires an explanation based on equipment, and thus, they are implemented in two distinct artefacts.

B. Demand-Side Management

Demand-side management measures of peak clipping and load shifting were shown to reduce billing costs without affecting service provision on an Indian university campus by Balasubramonian et al. [13], and Patil and Tanavade reported staged interventions such as lighting replacement and scheduling based switching [14], while Lieta. studied semester scheduling as an indirect demand response strategy [15]. These studies consider scheduling as a manual process of planning; here scheduling is computed from a forecast and priced out against alternative options.

C. Research Gap

While no step is not investigated separately and none of these is claimed to have any algorithmic innovation, the gap filled in this research lies in the lack of integration between the steps. The gap is in the way that analysis output gets turned into an institutional decision from forecast to residual to equipment to cost to ranked action. There are two other omissions present in the literature review, which appear repeatedly. In many energy analytics investigations, the financial impact is simplified down to an energy rate estimate, ignoring the demand charge part and thus the class of demand-side interventions which do not reduce consumption but peak. Also, data collection is usually considered a single unified instrumented feed source.

III. METHODOLOGY

The structure is designed in a way that forms a pipeline where each stage processes the output of the previous stage and gives a unique meaning. Figure 1 below shows the design.

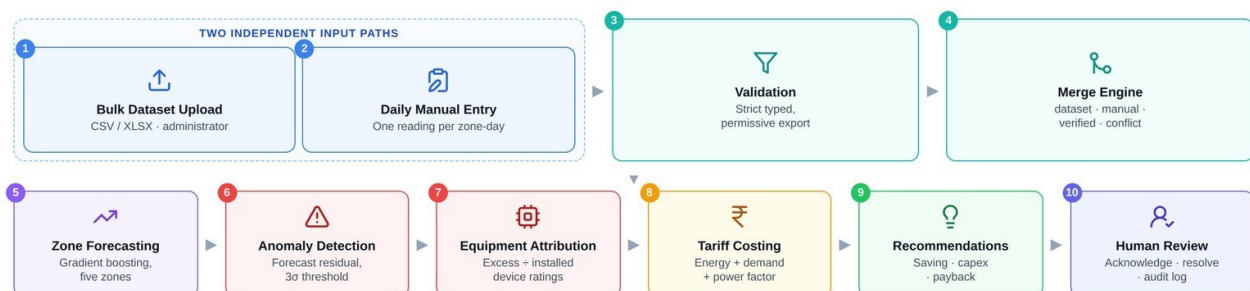


Fig. 1. The processing pipeline: multi-path acquisition, reconciliation, forecasting, detection, attribution, pricing and human review.

A. Study Site and Zone Partition

The unit of study is one unit of an institution in Chennai, segmented into five usage zones and not considered as a single meter since the rooms having the same usage pattern will have a similar consumption pattern. The physical audit of the building gives the connected load of the zones shown in Table I, which sets the upper limit of any reading of the zone. The consumption of the zones is highly uneven to an extent that two laboratories alone contribute nearly half the block consumption.

TABLE I. ZONE PARTITION AND AUDITED EQUIPMENT INVENTORY.

Zone	Rooms	Connected load	Share
Computer Labs	2	39.6 kW	48.7%
Classrooms	23	22.8 kW	29.1%
Server Room	1	6.2 kW	10.0%
Common Areas	5	2.8 kW	6.4%
HOD Rooms	3	6.9 kW	5.7%

B. Dataset Construction

Sub-metering was not possible and the institution's metering system provides monthly aggregation only. Instead of using a public dataset of a separate building, the time series was generated based on a census of the block's devices, a Chennai weather model and the institution's academic calendar. The hourly demand is determined as the product of installed device's power and a duty cycle depending on the hour, day and temperature:

$$E(z,t) = \sum_i P_i \cdot n_i(z) \cdot d_i(t, \theta_i, c_i) / 1000(1)$$

where d_i represents the duty cycle of type i device during time t when the temperature is equal to θ_i and day type is c_i . The duty cycle values are physically motivated as opposed to being estimated: as the temperature increases, the fan usage increases, lighting depends on occupancy with adjustments for daylight hours, laboratory computers are based on time schedules, and server load is constant. See table II below for input values for the generators of the series.

TABLE II. GENERATOR INPUTS FOR THE SYNTHESISED SERIES.

Generator input	Specification
Equipment inventory	5 zones, per-device rated power and counts
Weather model	Chennai seasonal and diurnal temperature, humidity
Academic calendar	2 semesters, 2 exam periods, 2 vacations, holidays, working Saturdays, event days
Duty-cycle rules	Per device type, conditioned on hour, day type, temperature
Anomaly categories	7 (overnight, weekend, vacation, holiday, baseline creep, equipment fault, meter dropout)
Anomaly injection	11 events, 532 affected hours, recorded in a withheld ground-truth file
Output	43,800 rows, 147,155 kWh, peak 162.7 kW, calendar year 2025

As the locations of the faults are known, the success of detection can be determined using ground truth. There are no labels on public consumption data sets, which is why qualitative detection performance is described in the majority of reviewed literature.

C. Multi-Path Acquisition and Validation

Data is acquired through three paths file uploads by an administrator, one daily total typed by a block manager,

and one hourly reading typed at the time the meter is read. The third path exists because a daily total doesn't show the time of consumption, and all equipment operating from 02:00 to 05:00 gets hidden in it. Physical limit for a zone depends on the inventory:

$$C_z = (\sum_i P_i \cdot n_i(z) \cdot 24 / 1000) \cdot \lambda(2)$$

In this case, $\lambda = 1.25$ is a headroom factor that provides for tolerance in the meter and non-metered equipment, hence, C_z is not a sum of the nameplate but rather a permissive limit. Validation is intentionally biased. An entered number greater than C_z is considered invalid, since if this number is higher than the capacity of the equipment in place, it is definitely a typing mistake. However, the very same number when received in the import file is considered valid and a warning flag is set for it because it could be a real problem - the one the detector was designed to detect. Rejecting it would suppress the evidence the system was built to surface.

D. Reconciliation Policy

TABLE III. RECONCILIATION POLICY APPLIED PER ZONE-DAY.

Condition	State	Value used
Dataset only	dataset	Imported series
Manual only	manual	Total, spread by profile
Both, within τ	verified	Manual total, dataset shape
Both, beyond τ	conflict	Manual, pending decision
Administrator ruled	resolved	Chosen source, audited

Where sources agree within tolerance the manual total is preferred for the daily quantity while the imported file

supplies the hourly shape. Disagreement beyond tolerance is not averaged away: it is recorded as a conflict and an administrator must rule.

E. Forecasting

A case-specific histogram gradient boosting regressor is created for each of the regions having different scale and dynamics. 38 characteristics are input into the regression model which are based on four types of calculations: autoregressive history lagged by 1, 2, 24, 48 and 168 hours, trends by moving averages with lagged shift, hourly and weekly cyclical coding, and weather data with thermal units associated. The models are constructed on the data from the period January to September and evaluated from the data from October to December that is not taken into account during the forecast.

An important feature to be considered in terms of the performance of the building is profile giving hourly average consumption per day teaching type that we have used only on the training data. When the period changed from the normal studying to exam sessions, every lagged parameter was from the usual working day and did not correspond to exam circumstances making evaluation especially difficult; profile supplied the right number regardless of the most recent information. The evaluation process is summarized in Table IV.

TABLE IV. FORECASTING CONFIGURATION AND EVALUATION SETUP.

Parameter	Value
Estimator	HistGradientBoostingRegressor
Models / features	5 (one per zone) / 38
Max iterations / depth	700 / 7
Learning rate	0.05

Train / test split	Jan-Sep / Oct-Dec, chronological
Baseline	Seasonal-naive, lag 168 h, recomputed

F. Detection and Equipment Attribution

An observation is flagged when the standardized residual exceeds a threshold k of the training standard deviation:

$$|r(z,t)| / \sigma_z > k, \quad r(z,t) = E(z,t) - \hat{E}(z,t) \quad (3)$$

Because the reference has already taken into account the weather, hour, and type of day, the residual is consumption not explained by the context of operation, such that a normal working Monday is not deemed anomalous just because it suffers from different behavior compared to the previous Sunday. Residual thresholding is not affected by a gradual slow drift of the baseline ingestion of any gradual increase in the registration baseline into the history features; therefore, implementation uses a rolling baseline approach in addition to the residual control.

Attribution means converting the residual into a physical assumption. For an excess ΔP in zone z , the number of devices of type i required to account for it is given by: Admissibility of that is defined as follows:

$$u_i = 1000 \cdot \Delta P / P_i, \quad \text{admissible iff } u_i \leq n_i(z) \quad (4)$$

Hypotheses that ask for more units than what is available are eliminated. A shortlist of what can be checked by the tech on location is what remains. A 19 kW overnight excess in the labs is roughly equal to around 127 desktop equivalents of 150 W, which is not equal to 240 installed units in contrast to ceiling fans that would require the installation of 253 units against 30 already in place.

G. Cost Model

The cost is calculated based on the high voltage tariff rather than a fixed tariff, using the prescribed indicative constants.

TABLE V. TARIFF CONSTANTS. VALUES ARE INDICATIVE PUBLISHED FIGURES, NOT THE INSTITUTION'S CONTRACTED RATES.

Constant	Value	Basis
Energy rate	₹8.05 / kWh	Indicative HT commercial
Demand rate	₹580 / kVA / month	Indicative HT commercial
Sanctioned demand	150 kVA	Site connection
Power factor / target	0.94 / 0.95	Modelled / tariff target

$$S_{\max} = P_{\max} / \cos \phi \quad (5)$$

$$C_{\text{annual}} = E_y \cdot c_e + 12 \cdot S_{\max} \cdot c_d \quad (6)$$

where E_y is the annual energy use (kWh), c_e the energy rate (₹8.05/kWh), $P_{\max}$ the measured peak active power (kW), $\cos \phi$ the power factor modeled (0.94), $S_{\max}$ the apparent power on which the demand charge is imposed (kVA), and c_d the demand rate (₹580/kVA/month). The apparent power is being used for the calculation of demand charge. Thus, the recorded peak of 162.7 kW turns out to correspond to

173.1 kVA based on the given power factor. The platform is designed in such a way that the power factor charge and statutory tax is applied but from (5) and what was

discussed in Section IV-D only the two main components are taken into account.

H. Execution and Assessment Approach

The setup is carried out as a service-oriented software utilizing a Python service layer, a relational database system with versioning migrations, and a dashboard for browser interface. The access here is differentiated by roles and processed on the server side, every operation they cause is monitored, and the implementation layer includes 371 automated tests of 90% statement coverage.

The method of evaluation was decided in advance. There is no shuffling or random partitioning of the time series since it can also lead to using future information to provide forecasts for the past. All the hours with abnormally behaved readings will be dropped out from the training of this model so residuals can be interpreted as deviations from normal working. The evaluation accuracy is implemented in MAPE and also in the form of mean absolute error calculations with the reason to have more accurate and objective informative results.

IV. RESULTS

A. Load Characterization

The load pattern throughout the working day consists of two peaks occurring at noon and again at 3 PM, separated by a drop due to lunchtime, corresponding to the two sessions conducted in the laboratories. The holiday load is approximately one-fifth of the teaching load, but it never drops to zero, while the whole system has a standing load of 4.5 kW which works without people present at 3 AM, meaning around 39,400 kWh for the year.

B. Forecasting Accuracy

TABLE VI. FORECAST ACCURACY BY ZONE. MADE IN kWh PER HOUR; NAIVE IS THE RECOMPUTED SEASONAL-NAIVE BASELINE MAPE; CUT = (NAIVE - MODEL) / NAIVE $\times$ 100.

Zone	MAPE	MAE	Naive	Cut
Computer Labs	5.6%	0.250	72.0%	92%
Classrooms	22.9%	0.733	137.3%	83%
Server Room	1.6%	0.023	3.3%	52%
Common Areas	4.3%	0.047	18.4%	77%
HOD Rooms	25.8%	0.102	36.0%	28%

Every region surpasses the baseline depicted in Figure 2, which shows the comparison on a logarithmic scale. Block-level accuracy from the offline evaluation shows 5.75 per cent MAPE, mean absolute error of 0.96 kWh per hour and the coefficient of determination of 0.987.

The disparity across zones calls for explanation and not bypassing it. HOD Rooms show 25.8 per cent but average only 0.29 kWh per hour, meaning that absolute error of 0.10 kWh per hour is of no practical significance and it shows high percentage only due to small denominator; the zone constitutes 5.7 per cent of consumption. Classroom error occurs during 07:00 and 17:00; these are transition hours when fan and lighting utilization is driven not by the schedule or temperature but behavior of the occupants.

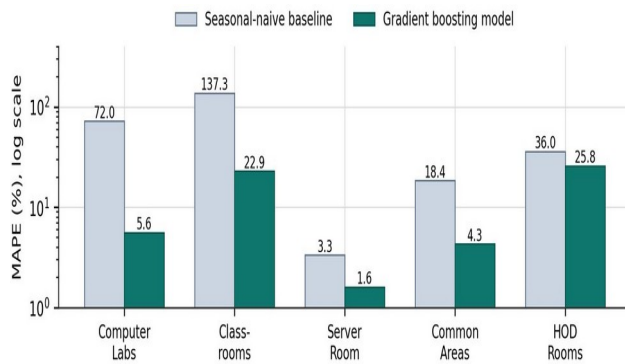


Fig. 2. Per-zone forecast MAPE against the recomputed seasonal-naive baseline, logarithmic scale, over non-anomalous test hours.

C. Detection Evaluation

Evaluation was carried out for the purpose of detection against the eleven labelled events using the reference predictor obtainable by recomputing from the available series: per zone mean conditioned upon day type and hour fitted according to the training hours of non-anomalous nature. This means that the predictor was employed instead of the training models because the service in operation produces the reference profile (Section V-C). The figures in this case reveal the characteristics of the reference predictor and should not provide an indication of the performance estimate of the trained models.

TABLE VII. HOUR-LEVEL DETECTION AGAINST GROUND TRUTH AT INCREASING THRESHOLDS, AND EVENT-LEVEL RECALL.

k	Precision	Recall	F1	Events
3	0.063	0.415	0.110	9 / 11
4	0.096	0.378	0.154	9 / 11
5	0.143	0.336	0.201	9 / 11
6	0.268	0.305	0.285	9 / 11
8	0.532	0.237	0.328	9 / 11
10	0.825	0.177	0.291	9 / 11

At every point of threshold studied, nine of the eleven cases appeared, while two exceptions were the overnight load event that occurred in the HOD Room as well as holiday consumption events that occurred in classrooms. The consumption level of zones that these cases appeared in was so low that the additive excess stays in normal residual spread. Hour-level precision is very weak with a value of 0.063 at the three-standard deviation threshold, and increases to 0.825 at the level of ten standard deviations, at which point the recall drops down to 0.177. Most of the false positives occur in Server room, classroom, and HOD Rooms because absolute consumption in the three zones is very low.

At the three-standard deviation threshold, nine out of the eleven cases were detected by the reference predictor. With the hour-level precision of 0.063, the operator would have to sift through approximately fifteen false alarms for every true case.

D. Cost Structure

TABLE VIII. MODELLED ANNUAL COST COMPOSITION USING THE INDICATIVE CONSTANTS OF TABLE V.

Component	Annual	Share
Energy (147,155 kWh at ₹8.05)	₹11,84,598	49.6%

Maximum demand (173.1 kVA at ₹580)	₹12,04,672	50.4%
Combined	₹23,89,270	100%

The single highest demand interval influences slightly more than half of the modeled yearly bill. The flat energy-rate model disregards the demand-charge component, and therefore it may produce a different ranking of interventions, since the measures to reduce consumption have an equal effect on the modeled bill by reducing 49.6 percent of the amount whereas the measures to reduce peak will act on the remaining 50.4 percent of the bill; hence the rankings were not calculated separately in this analysis. Peak coincidence analysis explains measured peak demand of 173.1kVA and the limit of 150kVA with the coincidence of laboratory activities during the hot afternoon. Postponing one laboratory session by an hour does not require any expenditures and does not affect consumption; however, it decreases the billed peak demand.

E. Objective Mapping

Table IX maps each objective to the evidence bearing on it and the outcome reached.

TABLE IX. OBJECTIVES MAPPED TO EVIDENCE AND OUTCOME.

Obj.	Evidence	Outcome
O1	Table VI, Fig. 2	Met; 5.6–25.8% MAPE by zone
O2	Section III-F worked example	Partially met; hypothesis set, not diagnosis
O3	Table VIII	Met; demand is 50.4% of modelled cost
O4	Table III, implemented policy	Met; five provenance states
O5	Tables VII and VIII	Met; baseline reproducible, detection weak

V. DISCUSSION

A. Interpretation

The outcomes reinforce the claim that, in this case, architecture and not algorithms is a reference. It is not the forecasting precision that produces a residual; it is a reference that makes the residual meaning-full. The residual becomes helpful when the inventory converts it to machines, the machine hypothesis when tariff priced it, and the price when it brings the result to the place of inaudible interventions. This can be illustrated by the demand-charge result: a constant energy-rate model does not take into account a demand component and therefore places a consumption-reducing measure higher than a peak-reducing measure, although neither of the implantation was measured separately.

The detection results put this into perspective. The pipeline works as intended and shows most of the events injected into it; however, the hour precision when using a reference predictor is too small and does not allow unsupervised working. The trained models have a lower forecasting error than the profile applied during the evaluation, but it has not been checked whether it allows achieving better detection results. A connection between the inference schedule and the path executed is the next engineering stage.

B. Comparative Analysis with Previous Research

In the context of research on prediction [1]–[6], this study essentially does not make any contribution; a block-level MAPE of 5.75% is unremarkable against earlier findings; therefore, no superiority of the study can be claimed. The contribution of the study in relation to the literature on detection [7]–[9] is in the findings rather than in the research itself. Compared to the body of work on explainability [10]–[12] this contribution is of ontological nature: whereas interpretability makes the model interpretable, inventory attribution solves the building.

In regard to the strength of the evidence, the comparison is unfavorable. The studies that work with metered data [5], [6], [13] assess outcomes compared to actual measurements that occurred and the faults that were available. The present research is compared to a synthetic process which includes faults intentionally added. Hence, the findings obtained cannot be compared on the quantitative level with those in the studies that used actual data. What the synthetic approach provides is certainty of ground truth which is missing in the studies.

C. What Is Not Demonstrated

- No verified energy or cost saving. No intervention was implemented and no post-intervention measurement taken. Reported savings are modelled consequences of modelled faults.
- No validation against metered data. Reported accuracy measures the models' ability to learn a physically motivated generative process.
- No confirmed attribution. The procedure narrows causes to a physically admissible set; it cannot establish which is correct.
- No live model inference in the deployed service. The forecast endpoint returns a day-type reference profile, because the trained models require a 38-feature matrix built offline. Reported model metrics come from that offline evaluation.

D. Threats to Validity and Limitations

As anomaly labels are based on preset failure patterns, construct validity is very important since unknown behaviors may go unrecorded. Internal validity seems to apply because of chronological data division and absence of abnormal hours in the training, though one may not notice some daily behavioral changes. However, external validity is conditioned by the use of a particular institutional block due to location-specific tariffs, inventory, and calendar values. Among other limitations is the use of estimated tariffs and only 11 modeled events, projected inventory values, and borrowed Canadian design ideas. Finally, to put the reproducibility of the results into practice, Table VI results were recalculated using the approach described in Section III-H.

VI. CONCLUSION AND FUTURE WORK

In this paper, a framework was discussed in which energy forecasting provides the base model, detection data is correlated with an audited equipment database and results are expressed in full high-tension tariff, so that measures are compared in monetary terms. While applying the synthesized year hourly data to the five-zones academic

block, several logarithmized tests were run for the model and 9 out of the 11 labelled faults were detected thanks to the reproducible reference tool which, however, was not perfect on the hour level, while combining the tariff calculation resulted in 50.4% of the annual costs being recognized as the highest demand cost.

Directly, two further steps follow. First of all, circuit-level metering would provide measured data for verification and the amount required to confirm the attribution, while reconciliation already has a metered input; hence no change is required..

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